

Day 6 Lab: Enriching Your API with External Data

Goal:

Today you will extend your existing API by integrating it with an external service. This will give your endpoints more value and demonstrate how APIs can be combined to create richer IoT or application experiences.

Your Task

You have two options. Choose one:

Option A: Enrich the Questions API with OpenAI

- Extend your Questions API so that when a client requests a question, your API also calls the OpenAI API to generate an *enriched answer* or *hint*.
- Example ideas:
 - Add a `/questions/{id}/hint` endpoint that uses OpenAI to generate a hint for the question.
 - Add a `/questions/{id}/explain` endpoint that asks OpenAI to explain the answer in more detail.

Option B: Enrich the Smart Home API with Weather Data

- Extend your Smart Home / IoT Devices API so that when a client requests a device, your API also calls the wttr.in API to fetch the current weather for the device's location.
- Example ideas:
 - Add a `/devices/{id}/weather` endpoint that returns the device info plus the current temperature and weather description.
 - Add a `/devices/{id}/enriched` endpoint that combines device data with weather data.

Requirements

- Add at least one new endpoint that combines your local data with external API data.
- Use Python's `requests` library to call the external API.
- Parse the response and merge it into your own API's response.
- Handle errors gracefully (e.g., if the external API is down, return a clear error message).

Hints (but not full instructions)

- Look up the documentation for the API you choose:
 - OpenAI API: <https://platform.openai.com/docs>
 - wttr.in: <https://github.com/chubin/wttr.in>
- You will need to figure out:
 - How to authenticate (OpenAI requires an API key).
 - What URL to call.
 - What format the response comes in (JSON, text, etc.).
 - How to extract the fields you need.